Managing the safety in smart buildings using semantically-enriched BIM and occupancy data approach

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Abstract: The Internet of Things (IoT) has transformed our ordinary environments into technologically-enhanced smart environments by seamlessly integrating the communication and computing capabilities of different building systems for understanding the building occupancy and adjusting the building environment accordingly while ensuring the occupants` safety. However, the sensory data of occupants get useless if it is not correctly mapped with its corresponding semantic information related to building locations for understanding occupant interactions using a building context. In the Architectural, Engineering, and Construction (AEC) industry, Building Information Modeling (BIM) is one of the preferred approaches to manage building data during a building lifecycle. Existing literature includes numerous solutions based on BIM and sensory data to manage building operations for a broad range of safety monitoring applications. However, BIM needs to track the evolution of contextual information associated with the building locations in terms of spatial (e.g. change in the room dimensions) and semantic changes (e.g. change in the functionality of a room). The latest spatial and semantic information of building locations with the former information will contribute to an improved understanding of building occupancy concerning the changes occurred in the building environment. This historicization of building information along with occupancy data will help Health and Safety (H&S) managers for conducting cause and effect analysis in case of building accidents to investigate what kind of change in the building has led to a safety hazard. To fill this research gap, a framework named *Occupant Behaviors in Dynamic Environments* (OBiDE) is used to construct a system for monitoring the occupancy behaviors in a building using historicized spatio-temporal data and semantically-enriched building information. The proposed system will provide a centralized knowledge base to study occupancy behaviors not only for safety monitoring but also for different facility management applications that deal with dynamic building environments.

Keywords: Behaviors; BIM; Dynamic facilities; Spatio-temporal trajectories; Safety management

1. Introduction

Buildings are one of the critical components of a human`s living environment (Jia et al. 2019). The idea of smart buildings initiated with the increase in the utilization of advanced Information and Communication Technologies (ICTs) to buildings and their linked systems (e.g. Heating, Ventilation, and Air Conditioning (HVAC)) so that the buildings and their occupants can be monitored and controlled remotely, and in a cost-effective way. The integration of ICTs with building systems is the prerequisite for achieving the realization of smart buildings. This includes deploying sensors in buildings for monitoring the occupant behaviors, processing and enriching collected occupants` data using state-of-the-art big data methods, cloud and fog computing for storing the acquired behaviors and performing different real-time data analytics, etc. (Jia et al. 2019). In the past decades, a huge number of studies have been conducted on smart buildings by incorporating the dynamicity of occupant behaviors for providing reliability, safety, and energy efficiency in building services to occupants without compromising their comfort level (Arslan et al. 2019a; Arslan et al. 2019b). Responding to occupant actions while they have been executed permits building supervisors and H&S managers to investigate their complete effects in the building environment. However, this process of analyzing the behaviors using real-time sensory data gets complex when the dynamicity (spatial and semantic changes) of the building facilities is included for occupant behavior understanding.

The building facilities are dynamic as they evolve spatially as well as contextually (Arslan et al. 2019a; Arslan et al. 2019b). Here, a context is any type of data that defines the building facilities. An ideal example of a dynamic building environment that evolves in both geometry and contextual information is the construction site. New structural and architectural supports are frequently added on worksites, while others are detached based on the construction work needs. Whereas, constructed facilities undergo less geometrical evolutions but many changes in the contextual information related to building facilities (Arslan et al. 2019c). For instance, an office in the building is now storage and has a different purpose (i.e. a semantic change). The spatial and semantic evolutions open more data management challenges to keep track of the spatial and semantic changes in the attributes of a building environment. Existing literature consists of many building monitoring solutions based on different ICTs for occupant behavior analysis (Ma and Wu 2020; Bonci et al. 2019). These systems are executed in real-time for sensory data collection and give timely notifications using handheld devices to building supervisors and H&S managers to take necessary actions in case of any abnormality observed in the occupant behaviors across building facilities. Existing literature suggests that, for studying the occupant behaviors, the occupancy is the fundamental and the pre-requisite to infer different occupant

activities (Wagner et al. 2018; Yan et al. 2015; Chen et al. 2015; Hong et al. 2015). For monitoring the building occupancy, there exist two methods for data collection which are; direct and indirect. The 'direct' method collects occupancy data using different ICTs (e.g. smart cameras, Radia Frequency Identification (RFID), beacons, cellular, etc.). Whereas, the indirect methods compute occupancy based on the data acquired using environmental monitoring sensors (e.g. Carbon dioxide), and stochastic modeling. To have more information on the methods for occupancy data collection, see the research of Yoo et al. (2020) and Zhao et al. (2014). Once the occupancy data is collected, it is enriched with the contextual information related to the building. The building information is typically extracted using an OpenStreetMap (OSM) or BIM-based data files (Arslan et al., 2019a). Though, the BIM approach is recommended as it holds the up-to-date building information throughout the building lifecycle (Arslan et al. 2019a). Existing literature contains numerous approaches based on building information and real-time sensory data for generating semantically-enriched BIM models to study occupancy behaviors for different safety management applications (as mentioned in Table 1). However, the dynamicity of building facilities while understanding the occupancy levels in a building for ensuring safety is not yet adequately explored in the existing literature. To address this gap, this study utilizes the framework *OBiDE* to model building occupancy for designing a safety management application in a dynamic building environment scenario (Arslan et al. 2019d).

The rest of the paper is ordered as follows: Section 2 describes the background literature of the presented study. Section 3 presents proposed system implementation in which the occupancy is measured using the spatio-temporal trajectories of occupants which were collected in a building using Bluetooth Low Energy (BLE) beacons. Secondly, the occupancy data is integrated with BIM software. Section 4 provides discussion and Section 5 presents a conclusion and some limitations of this study.

2. Background

Occupants perceive, move and perform actions in the buildings as well as interact with the building environment (other occupants and the building objects) for achieving their desired satisfaction (Wagner et al. 2018; Yan et al. 2015; Chen et al. 2015; Hong et al. 2015). The occupant actions are typically initiated from a series of entangled mental thinking which controls occupant actions and interactions with the building environment. These observable actions or interactions of occupants in response to external or internal stimuli are called occupant behaviors (Wagner et al. 2018; Yan et al. 2015). The understanding of the occupant behaviors implies inferring the drivers and goals using their

actions to identify their influence on the environment and thus to decide which actions are more suitable for effective building management. Responding to the occupant behaviors while they have been executed permit building analysts to study their effects on the building environment (Wagner et al. 2018; Yan et al. 2015). However, the consequences of their executed behaviors may result in serious problems while maintaining safety in buildings. To avoid such undesirable situations, a dynamic and real-time understanding of occupant behaviors is required while their actions are still evolving in the changing building environment.

Over the last decade, occupant behavior understanding has attracted the strong interest of the researchers for better monitoring and controlling the building facilities. To understand the occupant behaviors for different building management applications, in the existing literature (Wagner et al. 2018; Chen et al. 2015; Hong et al. 2015; Yan et al. 2015), an extensive range of different types of sensors (wired and wireless) for monitoring occupants and the environment are present. These sensors contribute towards the enrichment of environmental information for modeling occupant behaviors and their interactions (energy consumption, etc.) with the buildings. After collecting the required sensory data, this data is mapped with the corresponding building information to analyze the occupant behaviors using a building context (Yan 2011; Arslan et al. 2019a). To employ the building information, Building Information Modeling (BIM)-based platforms are utilized and preferred over traditional 3D Computer-aided design (CAD)-based systems in the AEC industry (Arslan et al. 2019b). A BIM model is a digital representation of the physical and functional characteristics of a facility (Arslan et al. 2019d). The BIM model provides a centralized source of shared knowledge of facility information for the building managers to use and maintain information throughout its lifecycle (NBS-US 2019). Each building component in a BIM model is a solid object containing its physical as well as the semantic information (i.e. alphanumeric properties) (NBS-US 2019; Liu et al. 2019). However, BIM models lack the feature of evaluating the current state of the building environment in real-time for building management processes (Stojanovic et al. 2018). Here, a state of a building object (e.g. a building facility) refers to its environmental data (e.g. temperature, humidity, carbon dioxide, etc.), the status of people inside the location (e.g. tracking, occupancy monitoring and identifying their activities), analyzing the utilization rate of building equipment or location, and performing localization and preventive maintenance of building objects for different BIM-based application scenarios (Tang et al. 2019). The process which adds meaningful information about states of building objects in a digital BIM model referred to semantic enrichment process (Sacks et al. 2017). Whereas, the resulted building model after

undergoing the semantic enrichment processes is called as a *semantically-enriched BIM model*. Also, for designing real-time building monitoring solutions using the BIM models, building objects (locations) needs to be constantly updated with the current state of the actual building environment (Stojanovic et al. 2018; Tang et al. 2019).

To incorporate the real-time sensory data of a building environment into a BIM model for studying occupant behaviors in the context of a building facility, wireless sensor technology has gained great importance in real-time monitoring of the buildings (Yan et al. 2015). ***"Some of the latest BIM and sensor-based integrated solutions for different building management applications from the literature are presented in Table 1"***. As it can be observed from Table 1 that the integration of BIM data with real-time sensory data acquired from different sensors has provided us a centralized digital building model to study different types of occupant behaviors for building management applications (Tang et al. 2019). However, the quality of acquired sensory data for the enrichment of BIM models differs significantly based on the resolution of the deployed sensors **"*see Fig. 1 below*"**.

Fig. 1 System resolution (Modified from Arslan et al. 2019c)

Table 1 BIM and sensor-based existing systems for facility management applications

No.	Use case	Data type	Key technologies / concepts
1	**BIM-based building fire emergency management** (Ma and Wu 2020)	Temperature, smoke and carbon dioxide data	Autodesk Revit software, SQL server and voice broadcast device.
2	**Stations-oriented indoor localization** (Yoo et al. 2019)	Wi-Fi signal strength	Dynamo, Autodesk Revit software and MySQL database.
3	**Infrastructure-free visual indoor localization approach** (Acharya et al. 2019)	Images	Cameras
4	**Wireless electric appliance control for smart buildings** (Rashid et al. 2019)	Location data	Ultra-wideband (UWB) anchors and real-time location tracking system (RTLS)
5	**A cyber-physical system approach for building efficiency monitoring** (Bonci et al. 2019)	Temperature and carbon dioxide data	Matlab, Autodesk Revit software and SQL Server
6	**Augmented reality system for facility management** (Baek et al. 2019)	Location data	Nvidia graphics processing unit (GPU)
7	**Detecting, identifying and localizing lighting elements in a building** (Troncoso-Pastoriza et al. 2019)	Lighting data	LiDAR sensors and an inertial measurement unit (IMU)
8	**Optimal path planning for dynamic building fire rescue operations** (Chou et al. 2019)	Location data	Bluetooth sensors
9	**Indoor thermal environmental design system** (Fukuda et al. 2019)	Temperature and humidity data	Autodesk Revit software and Augmented Reality (AR) devices
10	**BIM-sensor integration for responsive building management and operations** (Kazado et al. 2019)	Temperature and carbon dioxide data	Autodesk Revit software and Autodesk Navisworks software

11	**Integrating sensor, occupant and BIM data for building performance management** (Rogage et al. 2018)	Temperature and humidity data	SQL Server
12	**Building environmental monitoring under BIM environment** (Zhong et al. 2018)	Temperature, humidity, noise, carbon monoxide, carbon dioxide and methane data	SPARQL Protocol and RDF Query Language
13	**An IoT visualization BIM platform for decision support** (Chang et al. 2018)	Temperature, humidity and carbon dioxide data	Arduino, Raspberry Pi and Dynamo
14	**BIM-based indoor guidance system** (Ferreira et al. 2018)	Location data	Bluetooth Low Energy (BLE) beacons
15	**BIM-based system for building performance monitoring** (Kang et al. 2018)	Temperature, humidity and light data	DHT11 sensors and PT550 light sensors
16	**Government Open data and sensing data integration** (Lee et al. 2019)	Vibration and temperature data	MariaDB and Autodesk Revit software
17	**Real-time dust monitoring and visualization in BIM** (Smaoui et al. 2018)	Dust data	PMS5003 and PMS7003 dust sensors

The resolutions concerning spatial, temporal, occupant and level of semantic information are combined to determine the overall system resolution for capturing the occupancy data to study different occupant behaviors (Wagner et al. 2018; Yan et al. 2015; Melfi et al. 2011; Tang et al. 2018). The spatial resolution defines the size of targeted building area (i.e. buildings, floor, rooms, etc.) for data collection. Whereas occupant resolution typically has 4 different levels of information which are; 1) occupancy detection using 0 or 1 values, 2) counting occupants in a building facility, 3) recognizing the occupant identities using their profile information, and 4) distinguishing the occupants` activities based on the use case application. In addition, temporal resolution defines the smallest time period in which variations in spatial and occupant resolutions are reported by the installed sensors across building facilities (Arslan et al. 2019c). Lastly, the semantic resolution describes the level of detail of contextual information of a facility extracted from a BIM model is mapped with the spatial, temporal and occupant data. As the system resolution of sensors increases, the building facilities get smaller, the semantic information linked to the building locations gets more precise, occupants get more recognizable individually and hence, their activities can be inferred to study their behaviors for different applications (Arslan et al. 2019c). However, the process of understanding occupant behaviors gets more complex when the dynamicity of the environments is added into the behavior understanding process (Cruz 2017). The change in the purpose or the position of the locations in the buildings will result in different occupant behaviors (Yan et al. 2015). The updated location context with the previous contextual information is required to be captured for studying the behaviors of the occupants in detail in relation to the changes occurred in the building. The historicization of

building information in the form of a semantic BIM model enriched with the occupancy data will help building managers for conducting *cause and effect* analysis (Arslan et al. 2019b).

3. Proposed System

The six different processes ***"see Fig. 2"*** are used to develop the OBiDE framework ***"see Fig. 3"*** which is later used to design a prototype system application ***"see Fig. 4"*** which is based on;

3.1 Defining occupant behaviors using Drivers, Needs, Actions, Systems and States (DNASS) ontology: In *DNASS* ***"see Fig. 3"***, *Drivers* are the environmental factors that contribute to stimulating occupants to achieve the desired need. *Needs* can be physical or non-physical that must be fulfilled to achieve desired satisfaction. *Actions* are the interactions with the building environment for achieving the desired comfort. *Systems* correspond to the building environment and its equipment, and *States* define the classification of occupant activities (Hong et al. 2015).

Fig. 2 An OBiDE framework having 6 different processes

Fig. 3 Generalized OBiDE framework (Modified from Arslan et al. 2019d)

Fig. 4 Occupancy modelling use case using DNASS

3.2 Capturing occupants` data: As occupancy is the prerequisites for any type of occupant behavioral understanding, an OBiDE framework is based on capturing the spatio-temporal data of occupants in dynamic building environments. Around 200 BLE beacons ***"see Fig. 5"*** were mounted inside a university building ***"see Fig. 6"***. Due to the environmental conditions, if the beacons are relocated in the dynamic building facility, the positions of the beacons in the system to tag the building locations against each longitude and latitude pair values need to be updated accordingly. To acquire the locations coordinates (longitude and latitude pair values) of occupants, an Android-based application is used in the smartphone devices of the occupants. As an application launches in a device, it senses the neighboring beacons and picks the best three beacons` signals for performing the geo-localization. Based on the Received Signal Strength Indicator (RSSI) of the beacons ***"see Fig. 7"***, location coordinates are generated and stored in a document database i.e. Mongodb for further processing in R studio. The collected dataset contains 8426 spatio-temporal points of 11 occupants having a sampling rate of 5 seconds during different times in a 12 days' interval. The dataset and its description can be found in Arslan et al. (2019e).

Fig. 5 Bluetooth beacons

Fig. 6 University building

Fig. 7 BLE beacon RSSI strength

3.3 Preprocessing the occupancy data: Once the spatio-temporal data of occupants is captured in the form of trajectories, this data is filtered and preprocessed based on the use case application. Spatio-temporal data of occupants often suffer from noise and environmental interferences (Zheng 2015; Yan et al. 2013). To reduce the level of noise in the collected data, median filtering is performed as it is preferred for datasets having outliers with low deviations (Zheng 2015). ***"A processed trajectory sample is shown in Fig. 8"***.

3.4 Semantic enrichment and storing the occupancy data: After preprocessing the occupancy data, the data is contextually enriched with the corresponding building locations to add more meaning to the collected trajectories using the building information extracted from BIM and an OSM file ***"see Fig. 9 and 10"***.

Fig. 8 Trajectory sample of collected BLE data

"Table 2 describes the mapping information for trajectories". Later, these spatio-temporal movements are stored in the STriDE data model. To have more details on the STriDE model, see the research of Cruz (2017).

Table 2 Mapping of BLE data with the building information

Location coordinates [Lon, Lat]	Geometry type	Floor	Room ID	Building Name
{[5.068580575, 47.31096787], 5.068377149, 47.31097487], 5.068814692, 47.31107513], 5.068567797, 47.31125648], 5.068565154, 47.31125453]}	Polygon	2	Office2	IUT

Fig. 9 Semantically-enriched spatio-temporal trajectory of an occupant having multiple timeslices

Fig. 10 The description of a timeslice from an occupant trajectory

3.5 Modeling occupancy behaviors: Based on the use case application requirements, occupant behavior modeling techniques are applied to the processed spatio-temporal occupancy data along with the supplementary data of the occupants and the building environment for achieving the desired classification of the occupant behaviors. There exist probabilistic and non-probabilistic approaches in the literature (Wagner et al. 2018; Yan et al. 2015; Chen et al. 2015; Hong et al. 2015) for occupant behavior modeling. To show a proof-of-concept application of modeling occupancy behaviors of occupants, the Hidden Markov Model (HMM)-based probabilistic method is chosen. The HMMs describes the occupant trajectories which have been captured from the BLE beacons as a series of Markovian stochastic processes where the probability distribution of a future state (i.e. low occupancy, in-range occupancy and high occupancy) of a stochastic (i.e. a random) process (a spatio-temporal processed occupant trajectory) at a building location of a user exclusively depends on its present occupancy state. This process of inferring the occupancy states using HMMs disregards the requirements of incorporating the historical states and eventually minimal training data

is required for modeling using HMMs. Generally, an HMM (λ) is defined using three parameters which are, transition probability matrix (A), emission probability matrix (B) and initial state probabilities (π). To train the HMMs for each building location, the Baum-welch algorithm is used which tunes the model parameters (A, B, π) for maximizing the probability of the observation sequences given by the model. The initial probabilities of the hidden states which are; 1) low occupancy, 2) in-range occupancy and 3) high occupancy are assigned an equal probability of occurrence i.e. $\pi = \{\pi_1, \pi_2, \pi_3\} = \left\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right\}$. For configuring the 'in-range occupancy' and 'high occupancy' states` values, the CLG Guide for Fire Safety Risk Assessment in Small and Medium Places of Assembly (GOV-UK 2006) is used. This Guide entails a method to keep the number of occupants in a building optimal to ensure safety in buildings. It defines a formula (i.e. number of occupants = Area of building floor (m²) / Occupant density) to calculate the number of occupants who can safely reside in the building by considering its functional attributes. To input the occupant density, the Guide mentions the occupant density in the case of the offices is 6.0m²/occupant. The working example is explained below;

Fig. 11 Calculating the occupant density in a building location

For a more comprehensive overview of the literature on occupant density, the reader is referred to Shen et al. (2014) and GOV-UK (2006). After training the HMMs for each building location, occupant trajectories are categorized into three states which are S_1 , S_2 and S_3 using the Viterbi algorithm. The Viterbi algorithm generates the most probable sequence of states which have produced the observations under the trained HMM model by implementing a global decoding mechanism. ***"As an example, Fig. 12 shows the categorization of a trajectory sequence of an occupant".***

Fig. 12. Decoded sequence states (S_1 , S_2 and S_3) of an occupant trajectory

3.6 Visualizing the classified occupancy behaviors: After categorizing the occupancy states, the latest occupancy states ***"see Fig. 12"*** are imported into an open-sourced BIM software (i.e. Autodesk Revit). The BIM model and its objects (building locations) are enriched with their corresponding states having occupancy information. For generating the occupancy visualizations, 'Dynamo' (i.e. a Revit plug-in) is used. Dynamo offers to program different pieces of logic with the help of a graph (i.e. connecting 'nodes' with 'wires') using a visual interface by reducing the need for

extensive coding. More information on Dynamo can be found in its user manual (Dynamo 2019). Dynamo graph is constructed to extract the building locations from a model that were labeled as 'rooms'. Later, this list of building rooms is enriched with their corresponding occupancy states for generating visualizations on a BIM software ***"see Fig. 13"***. The color scheme designed for the occupancy visualizations on a BIM model is for the sake of demonstrating the functionality of the developed system by showing the building locations by changing their colors. However, it can be changed based on user preferences.

Fig. 13. Visualizing occupancy on a building model in Revit software

4. Discussion

The main research question which guided this research was, how to establish an understanding of the occupancy in a dynamic building environment? The research question was motivated by the fact that occupant behaviors (i.e. occupancy in our case) evolve as well as the buildings. Existing systems for occupant behavioral understanding do not incorporate the information of dynamic environments where the building objects (occupants and building locations) evolve in geometry or related semantic information. The change in the semantics of building locations occurs more often in constructed facilities whereas, the spatial changes take place rarely in a building lifecycle. Such changes in the building need to be incorporated in the occupancy behavior understanding process as a change in the purpose or a position of building locations will result in different occupancy behaviors which ultimately represent different occupant activities. For instance, an occupant in a room labeled 'Office' is identified as a worker. Whereas, over time, due to the change in the purpose of the location, that 'Office' is now a 'Storage room'. The system should identify the same occupant as an 'intruder' if he does not have access to this location. But this will not happen as the system cannot historicize the details of semantic evolutions of a building environment. The updated spatial and semantic information related to the building locations with the previous information will contribute towards improved occupant behavioral understanding relating to different changes that occurred in the building. To address this requirement of monitoring occupancy in dynamic building environments that contain evolving building objects, a prototype system is designed which; 1) defined the required occupant behaviors using Drivers, Needs, Actions, Systems and States (DNASS) ontology, 2) captured spatio-temporal data of occupants using BLE beacons 3) preprocessed the occupancy data, 4) performed semantic enrichment to add building context to occupancy data and storing it into the STriDE model to maintain the historicization of occupancy in context of a building environment, 5)

categorized the occupancy into 3 levels (low occupancy, in-range occupancy, and high occupancy) using HMMs-based probabilistic model and 6) visualizing the occupancy using BIM software.

For studying the occupancy behaviors, BLE beacons were chosen for spatio-temporal data acquisition. The choice of BLE beacons was made because of its low cost of deployment and availability for the application. Based on the proximity analysis done by the BLE beacons, locations (i.e. longitude and latitude pair values) of occupants were captured and processed. The BLE beacons have provided the proximity of the occupants in a building. However, the proximity is not the same as the exact locations of occupants (i.e. absolute values of geographical coordinates). For developing the prototype system, the high precision in the occupants` proximity was not required as the trajectory data was needed for detecting the occupancy at the room level rather than capturing the exact positions of occupants inside rooms. Later, the occupancy behaviors were modeled using HMMs to categorize the occupancy of a building into 3 levels (low occupancy, in-range occupancy, and high occupancy). The BIM-based visualization is generated after enriching the BIM model with the occupancy information, which can help building managers to quickly analyze occupancy levels of different rooms not only for ensuring safety in a building but also support them for optimizing the operational efficiency (see Table 2 below). Also, the developed system will allow helping in identifying the present occupants across rooms and assist them to analyze the historical room's utilization over time.

Table 2 Potential applications of occupancy monitoring in Building management (Occupancy monitoring, 2019)

No.	Building management areas	Potential use cases of occupancy monitoring
1	**Health and Safety (H&S)**	Occupancy data providing the occupants` presence can be used by the emergency crew members to identify the occupant locations in case of an accident.
2	**Operation and Maintenance (O&M)**	Occupancy data indicating which building facilities have been used and therefore may need maintenance. Also, maintaining the building resources based on the occupancy of the building spaces.
3	**Security**	Triggering the alerts to building supervisors and H&S managers in case of unauthorized personnel occupancy in restricted areas of a buildings.

5. Conclusion

Occupant behavior is an important factor in attaining building performance goals that involve around H&S, O&M, and security. The implementation of smart buildings has integrated building systems using different sensors and actuators for optimizing the building performance based on occupant behaviors. However, if the occupant behaviors

(i.e. occupancy in our case) captured from the sensory data are not adequately mapped with the updated contextual information of a building. This will lead to inefficient utilization of building resources and eventually degrades the overall building performance. To study the occupant behaviors by enabling the historicization of spatial as well as contextual evaluations of a building environment, a prototype system is presented for analyzing occupancy behaviors using BIM software. One of the limitations of the developed system is that the study is executed in an already constructed building and involves a limited amount of semantic information of a facility for a BIM model enrichment. Additional work is required to include more semantic information from different data sources and to empirically test the system application on the dynamic building environments where spatial and contextual evolutions occur more often. This will help to understand the occupancy across different building areas during different times of the day. At the occurrence of an accident or an unauthorized personnel occupancy, alerts can be generated to building managers to reduce potential safety accidents. For the future direction of this research, it needs to be mentioned that currently this research is solely based on ICT concepts that were utilized based on the spatio-temporal data processing for occupancy monitoring application. This study presents a baseline model to monitor occupancy which is regarded as the prerequisite of occupant behavior understanding. To infer occupant activities in the building environment, a multidisciplinary approach is needed for developing advanced behavioral models. And this can be achieved by combining the knowledge of other than occupancy-related behavioral, social and psychological factors that contribute to occupant behaviors over time.

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